

VeriFrame - Simple Documentation

Professional Video Authenticity Detection Platform

Problem

In an era of rising digital manipulation and highly realistic deepfakes, determining the authenticity of video media is increasingly difficult. Fake videos are created to spread misinformation, manipulate elections, and damage reputations. Journalists, law enforcement, organizations, and everyday users need a quick, reliable way to verify if a video has been modified.

Solution

VeriFrame offers an advanced deep learning forensics platform to detect video manipulations. Users can upload any video file through a user-friendly drag-and-drop interface, track the analysis process in real-time, and get detailed visual and statistical breakdowns of the video's authenticity.

Platform Features & Sections

1. Landing Page:

The entry point introduces the platform. It features a cinematic hero section, animated highlights explaining the tool, a step-by-step walkthrough of the analysis process, and an FAQ. No login required.

2. User Login:

Uses secure, password-free Internet Identity built into the Internet Computer. Users log in once to gain access to protected features like video uploading and the analysis history dashboard.

3. Video Upload & Live Tracker:

A clean drag-and-drop area for uploading files. A real-time tracker displays the progress of the frame-level deep learning pipeline analysis.

4. Forensic Results View:

Once analysis completes, the results page displays:

- Authenticity Score (color-coded: Green = authentic, Red = manipulated)
- Deepfake Probability Percentage
- Frame Heatmaps showing suspicious localized regions
- Temporal Consistency Graphs mapping video timeline irregularities
- Detailed metric-by-metric breakdowns

5. Video History Dashboard:

A personal list of all previously analyzed videos. Users can search, filter, and export the database as a CSV report.

6. Reports & Sharing:

Analysis results can be downloaded as a PDF report or shared via a read-only secure link.

Demo Credentials

To test the application, use the default credentials below:

- Username: demo
- Password: demo

Note: You can also register a new account on the registration page.